

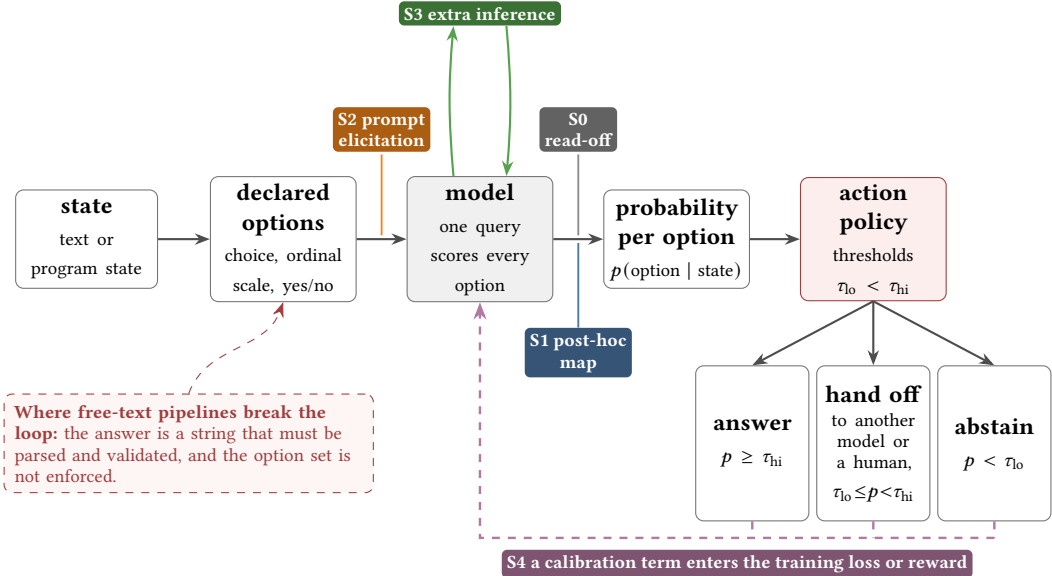


Fig. 10. **The calibrated decision loop, and where each stage injects calibration.** A decision model receives a state and a *declared* option set, scores every option in one query, and hands a probability per option to an explicit action policy. The policy drawn here answers when the probability is high, hands a less certain case off to another model or a human, and abstains when the probability is low; Table A1 lists the other low-confidence actions (trigger retrieval, filter the output, adapt compute, return a prediction set). The five stages of the taxonomy differ in where calibration enters: the model’s own read-off (S0), a post-hoc map (S1), a prompt that asks for confidence (S2), extra inference such as sampling or a conformal set (S3), or a calibration term in the model’s own training loss or reward (S4). The dashed callout marks where free-text generation pipelines break the loop (Table A1). Hosted and open typed-decision models (Tables 1 and 2) expose this interface: a state, declared options (a choice among labels, an ordinal scale or yes/no) and a probability per option.